Results

Machine-learning analysis of blood metals and metabolic syndrome in NHANES 2007–2018

1. Study population

Of 59,842 participants sampled across the six NHANES cycles, 11,144 adults met all inclusion criteria (Figure 1), and were divided into 8,915 training and 2,229 held-out test participants. Metabolic syndrome was present in 5,163 participants, a crude prevalence of 46.33%. Of these, 1,201 reached the three-component threshold only because the drug-treatment clause scored a component positive that the measured value alone would not have, and would have been classified as non-cases under a measured-value-only definition (35.55% prevalence).

Survey-weighted characteristics by metabolic syndrome status are given in Table 1. Participants with metabolic syndrome were on average 13.4 years older and had a body mass index 5.5 kg/m² higher. Among the blood metals, lead and selenium were higher in the metabolic syndrome group and mercury and manganese were lower, while cadmium did not differ appreciably; every difference reaching significance was small relative to the spread of the measurements.

Table 1. Survey-weighted characteristics of the analytic cohort by metabolic syndrome status. Continuous variables are weighted means (standard error); categorical variables are weighted percentages (standard error). Standard errors account for the NHANES complex survey design via Taylor-series linearization with domain estimation.

Characteristic	No MetS	MetS	p
Age (years)	41.82 (0.333)	55.27 (0.319)	< 0.001
Body mass index (kg/m²)	26.53 (0.123)	32.01 (0.183)	< 0.001
Blood lead (µg/dL)	1.33 (0.040)	1.48 (0.035)	0.007
Blood cadmium (µg/L)	0.47 (0.011)	0.50 (0.011)	0.108
Blood mercury, total (µg/L)	1.54 (0.050)	1.37 (0.045)	0.009
Blood selenium (µg/L)	194.76 (0.806)	197.95 (1.300)	0.037
Blood manganese (µg/L)	10.02 (0.079)	9.68 (0.092)	0.005
Income-to-poverty ratio	3.02 (0.043)	2.94 (0.051)	0.209
Sex: Male	48.70 (0.814)	50.19 (1.063)	0.125
Sex: Female	51.30 (0.814)	49.81 (1.063)	0.125
Race/ethnicity: Mexican American	9.30 (0.818)	8.36 (0.860)	< 0.001
Race/ethnicity: Other Hispanic	6.58 (0.620)	5.39 (0.572)	< 0.001
Race/ethnicity: Non-Hispanic White	64.86 (1.394)	70.37 (1.570)	< 0.001
Race/ethnicity: Non-Hispanic Black	10.85 (0.784)	9.29 (0.734)	< 0.001
Race/ethnicity: Other/Multi-Racial	8.41 (0.608)	6.59 (0.558)	< 0.001
Education level: <9th grade	4.69 (0.345)	7.15 (0.495)	< 0.001
Education level: 9-11th grade	9.59 (0.574)	12.21 (0.746)	< 0.001
Education level: HS grad/GED	20.97 (0.829)	25.83 (0.982)	< 0.001
Education level: Some college/AA degree	30.13 (0.955)	31.06 (1.111)	< 0.001
Education level: College grad or above	34.62 (1.395)	23.76 (1.154)	< 0.001

Characteristic	No MetS	MetS	p
Smoked ≥100 cigarettes in life: Yes	41.82 (1.007)	49.31 (0.993)	< 0.001
Smoked ≥100 cigarettes in life: No	58.18 (1.007)	50.69 (0.993)	< 0.001

MetS, metabolic syndrome. Unweighted group sizes: no MetS n = 5,981; MetS n = 5,163. Selenium and manganese were assayed only from the 2011–2012 cycle onward, so both are based on n = 3,382 without and n = 2,935 with metabolic syndrome. Metal concentrations use the cycle-appropriate laboratory weight; all other rows use the examination weight. *p-values for categorical variables are descriptive chi-square tests and are not design-corrected.*

2. Model performance

All eleven models were evaluated once on the held-out test partition (Table 2, Figure 2). CatBoost achieved the highest area under the receiver operating characteristic curve at 0.857 (0.840–0.872), and Naive Bayes the lowest at 0.730 (0.709–0.752). The apparent ordering at the top of the table does not survive sampling variability: in the paired bootstrap, LightGBM, XGBoost and Gradient Boosting all had AUROC differences from CatBoost whose 95% confidence intervals included zero ($p = 0.732$, 0.268, 0.172; Figure S2), so these four gradient-boosted ensembles form a tied leading group rather than a ranking. Every remaining model differed from CatBoost at $p < 0.001$. Two caveats apply: the reference was chosen as the test-set maximum, so the differences carry a winner's-curse bias and the p-values are anticonservative, and no adjustment was made for the ten simultaneous comparisons. The margin over the simplest baseline is modest in any case: logistic regression reached 0.845, only 0.012 AUROC units below CatBoost (0.006–0.018).

Calibration of the four leading models was close to ideal, with slopes from 1.06 to 1.10 against an ideal of 1.00 (Figure S3). Calibration did not track discrimination across the wider field, which AUROC alone would conceal: Random Forest reached a respectable 0.845 but a slope of 1.40, and the multilayer perceptron (0.43) and Naive Bayes (0.18) were worse in the opposite direction, producing probabilities far too extreme for the evidence behind them. Any of these could rank participants acceptably while giving an individual a badly wrong absolute risk. Decision-curve analysis (Figure S4) showed the leading models above both reference strategies across the range of thresholds examined; at a threshold of 0.20, CatBoost yielded a net benefit of 0.376 against 0.329 for treat-all, equivalent to identifying 47 additional true cases per 1,000 adults screened without treating more people unnecessarily. Threshold-level classification metrics are given in Table S4.

Table 2. Discrimination and calibration on the held-out test partition (n = 2,229), ordered by AUROC. Differences are against CatBoost from a paired bootstrap of 1,000 resamples in which the same resampled participants were scored under both models. **Models in bold are statistically indistinguishable from CatBoost.**

Model	AUROC (95% CI)	AUPRC	Brier	Cal. slope	Δ AUROC vs CatBoost (95% CI)	p
CatBoost	0.857 (0.840–0.872)	0.811	0.1534	1.06	reference	—
LightGBM	0.856 (0.840–0.871)	0.815	0.1541	1.10	0.0006 (−0.0026 to 0.0040)	0.732
XGBoost	0.855 (0.839–0.870)	0.809	0.1544	1.09	0.0017 (−0.0013 to 0.0046)	0.268
Gradient Boosting	0.855 (0.839–0.870)	0.804	0.1544	1.06	0.0021 (−0.0010 to 0.0052)	0.172
SVM	0.846 (0.829–0.862)	0.805	0.1594	0.99	0.0104 (0.0049 to 0.0153)	< 0.001
Logistic Regression	0.845 (0.828–0.861)	0.788	0.1609	1.08	0.0119 (0.0056 to 0.0181)	< 0.001
Random Forest	0.845 (0.827–0.861)	0.796	0.1629	1.40	0.0120 (0.0075 to 0.0165)	< 0.001
Decision Tree	0.830 (0.811–0.846)	0.769	0.1676	0.92	0.0267 (0.0185 to 0.0350)	< 0.001
MLP	0.806 (0.788–0.823)	0.761	0.1957	0.43	0.0506 (0.0385 to 0.0622)	< 0.001
KNN	0.802 (0.782–0.821)	0.738	0.1847	1.09	0.0548 (0.0434 to 0.0655)	< 0.001
Naive Bayes	0.730 (0.709–0.752)	0.669	0.2651	0.18	0.1266 (0.1082 to 0.1432)	< 0.001

AUROC, area under the receiver operating characteristic curve; AUPRC, area under the precision–recall curve; Cal. slope, calibration slope (1.00 = perfect). Lower Brier scores indicate better probabilistic accuracy.

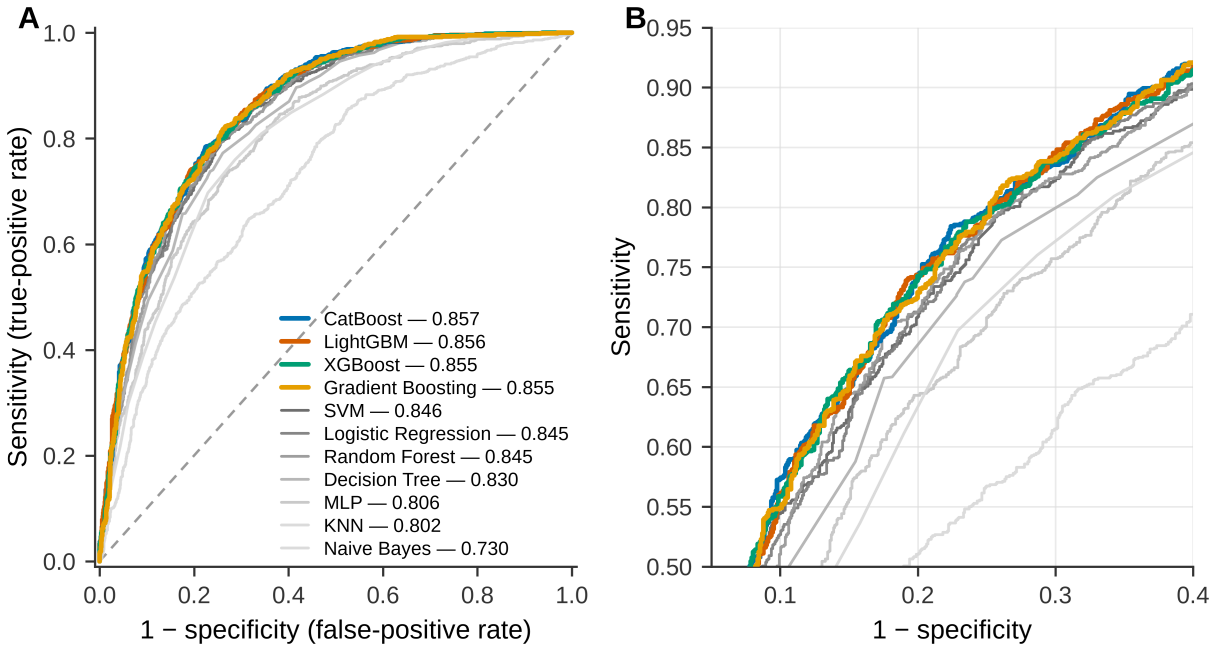


Figure 2. Receiver operating characteristic curves on the held-out test partition. **(A)** All eleven models, with AUROC in the legend; the four models statistically indistinguishable from the best performer are drawn in color and the remainder in grey. **(B)** Detail of the region of clinical interest.

3. Predictor importance

Body mass index and age dominated both importance analyses (Figures 3 and 4; full ranking in Table S5). By mean absolute SHAP value, body mass index (1.025) and age (0.991) were followed by vigorous recreational activity at 0.194, a fivefold drop from second to third. The gap was starker by permutation importance, where removing body mass index cost 0.160 AUROC

units and age 0.156, against 0.0084 for the third-ranked predictor. Between them these two account for nearly all of the model's discriminatory capacity; everything else adjusts risk at the margin.

The two methods agreed only moderately below the top of the ranking (Spearman $\rho = 0.62$ across 33 predictors; Figure 4A). Disagreement of this kind is expected rather than anomalous: mean absolute SHAP measures how far a predictor moves individual predictions, whereas permutation importance measures how much ranking performance degrades when that predictor is destroyed, so a predictor substitutable by correlated predictors scores well on the first and poorly on the second. Reporting either alone would have supported a confident but method-dependent account of which exposures matter. Two missingness indicators also carried non-trivial SHAP weight (whether average daily alcohol consumption and whether the income-to-poverty ratio were recorded), which are artifacts of questionnaire structure rather than biological signal and are excluded from the predictor ranking.

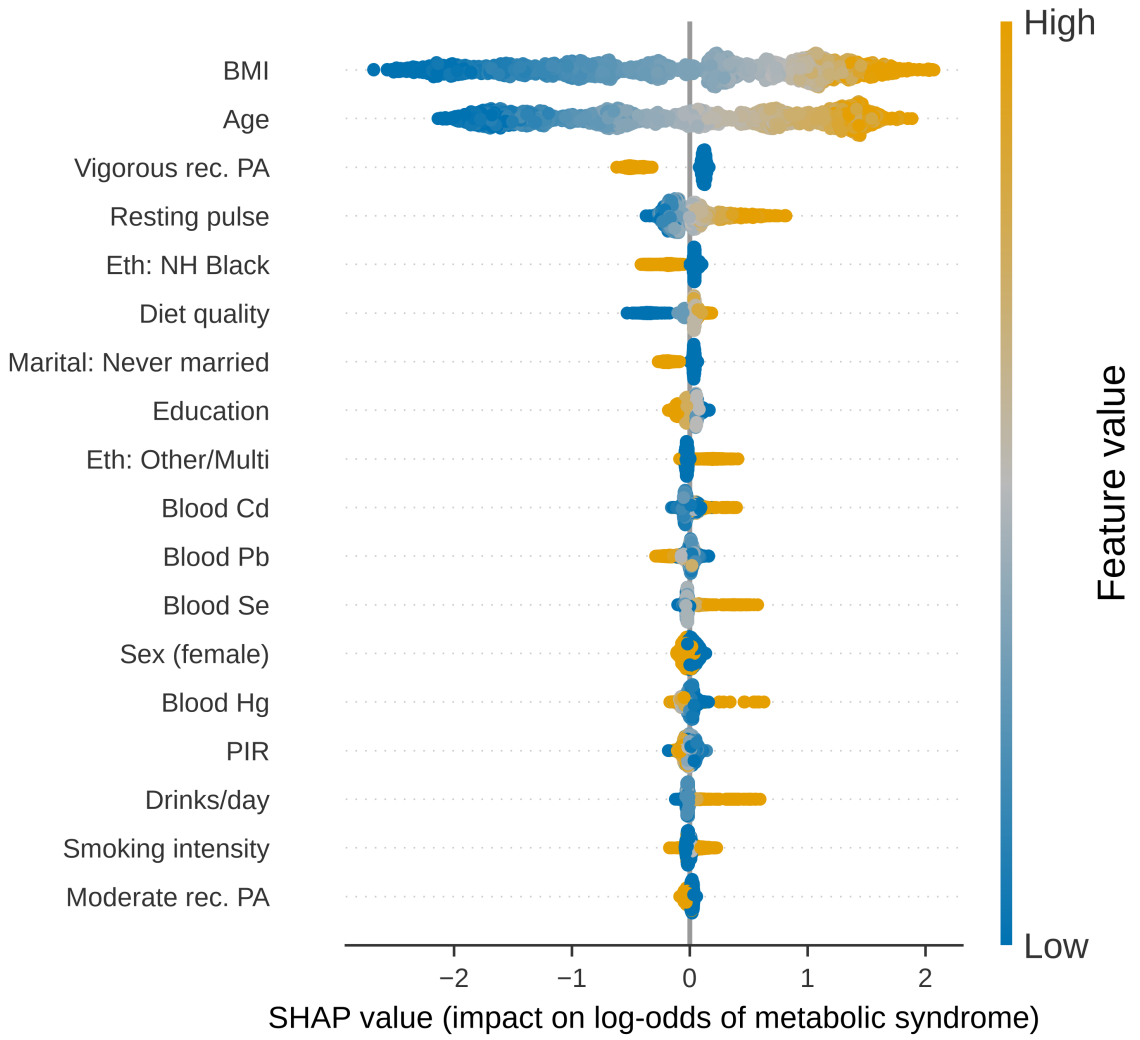


Figure 3. SHAP summary plot for the CatBoost model on the held-out test partition, showing the 18 highest-ranked features. Each point is one participant; horizontal position is that feature's contribution to that participant's predicted log-odds of metabolic syndrome, and color encodes the feature's value from low (blue) to high (orange). Abbreviations: BMI, body mass index (kg/m²); Vigorous rec. PA, vigorous recreational activity; Resting pulse, resting pulse (60 sec); Eth: NH Black, race/ethnicity: Non-Hispanic Black; Diet quality, self-rated diet quality (1=excellent...5=poor); Eth: Other/Multi, race/ethnicity: Other/Multi-Racial; Blood Cd, blood cadmium (µg/L); Blood Pb, blood lead (µg/dL); Blood Se, blood selenium (µg/L); Blood Hg, blood mercury, total (µg/L); PIR, income-to-poverty ratio; Drinks/day, avg. alcoholic drinks/day (past 12 mo); Smoking intensity, current smoking intensity (0=none, 1=some days, 2=every day); Moderate rec. PA, moderate recreational activity.

4. Blood metals

All five metals ranked in the middle or lower portion of the ordering, well below body mass index and age but above several demographic and lifestyle covariates. Their relative standing depended sharply on which importance method was used, and that method-dependence is itself the substantive observation; the precise ordering it produces is not. Cadmium was the highest-ranked metal by SHAP (rank 10 of 33) but fell to 18 by permutation importance, a gap of 8 places; the widest gap among the metals belonged to mercury (14 versus 27). Lead showed the

opposite pattern, ranking 11 and 9: close agreement between methods, and the only metal of the five in the permutation-importance top ten.

Agreement across independently fitted models tells the same story more decisively (Figure 4B). Lead was the most stable, ranking 4–9 across the four leading models and appearing in the top ten of every one. Selenium ranked 11–27, cadmium 17–29, manganese 19–29 and mercury 15–33, the widest swing of any predictor examined. Averaged across models, neither cadmium nor mercury nor manganese placed among the fifteen highest-ranked predictors. The permutation analysis also carries its own uncertainty, and reading the metal ordering without it would overstate what the data support: across ten permutations only lead (0.00123 ± 0.00044) and selenium (0.00069 ± 0.00023) had mean importances exceeding their own standard deviation, whereas for cadmium, manganese and mercury the mean was smaller than the run-to-run variability. The defensible statement is narrower than a rank list: lead and selenium carry permutation signal distinguishable from noise, and the other three do not.

The shape of each metal's contribution is shown in Figure 5. Selenium was the only one to rise monotonically across essentially its whole range, moving from −0.039 in the lowest bin, crossing zero at approximately 198 µg/L and reaching 0.235 in the highest, the largest positive contribution of any metal. Cadmium was U-shaped, contributing positively at both the lowest (0.050) and highest (0.104) concentrations with a trough of −0.049 between. Lead ran opposite to the conventional expectation, contributing positively only up to about 1.7 µg/dL and then declining to −0.151 in the highest bin. Mercury declined similarly, and manganese was flat throughout, every binned mean falling between −0.026 and 0.043.

The inverse gradient for lead deserves emphasis rather than quiet omission, because it runs against the hypothesis that motivated the analysis. In a cross-sectional model adjusted for body mass index, age and socioeconomic position, higher blood lead was associated with a lower predicted probability of metabolic syndrome. This is an associational pattern within one fitted model, not an estimate of a causal effect; residual confounding by smoking, which raises blood cadmium and lead while being associated with lower body mass index, is one plausible explanation among several, and adiposity-related dilution of blood metal concentrations is another. Disentangling these is what the Mendelian randomization component is designed to do, and the direction observed here should be read as a question posed to that analysis rather than an answer from this one.

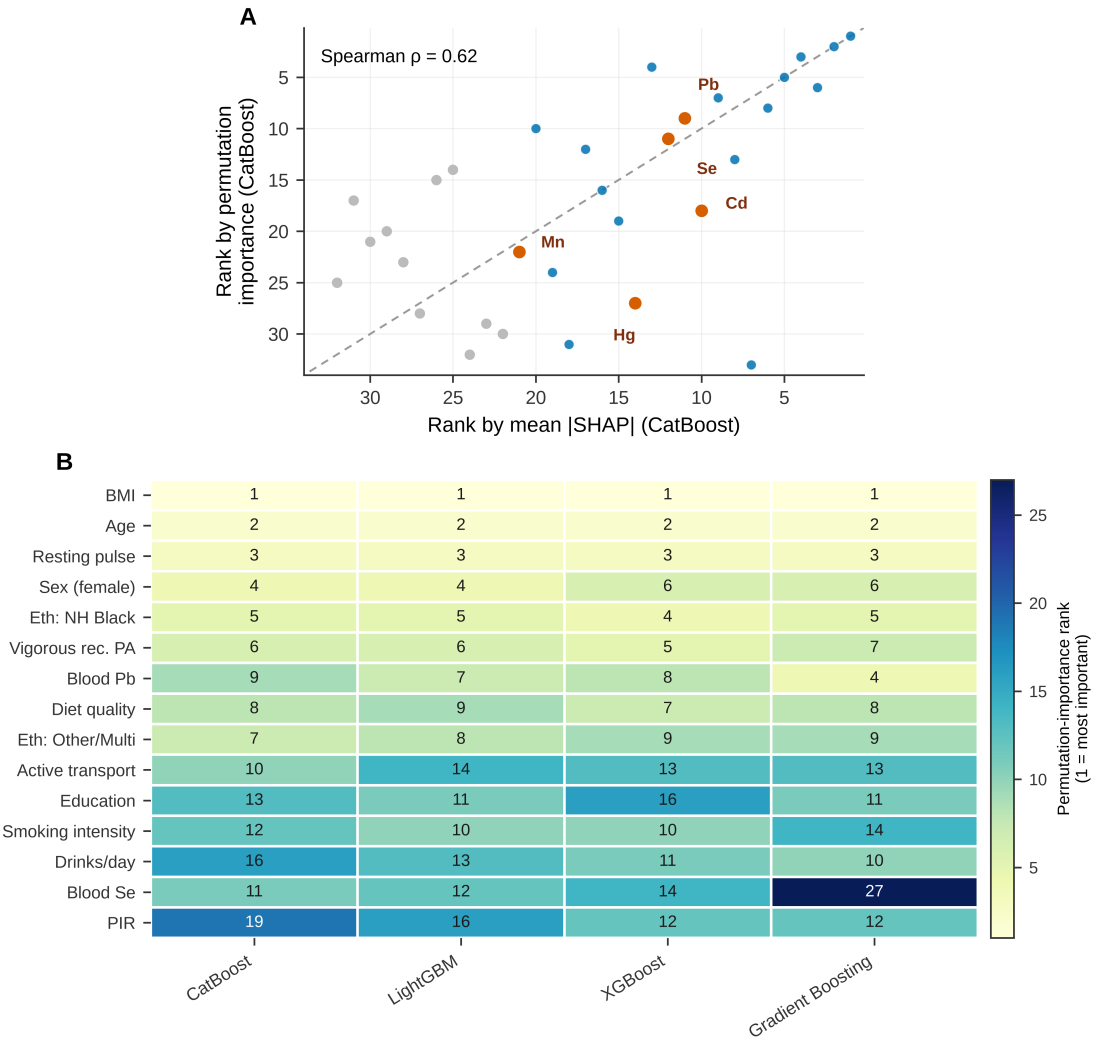


Figure 4. Agreement between importance methods and across models. **(A)** Rank by mean absolute SHAP value against rank by permutation importance for the CatBoost model; the five blood metals are highlighted and the diagonal denotes perfect agreement. **(B)** Permutation-importance rank of the fifteen highest-ranked predictors in each of the four leading models, fitted independently. A predictor whose row is uniform is important as a property of the data; one whose row varies is important as a property of a particular fitted model. Abbreviations as in Figure 3, plus Blood Mn, blood manganese (µg/L); Active transport, minutes walking/biking for transportation.

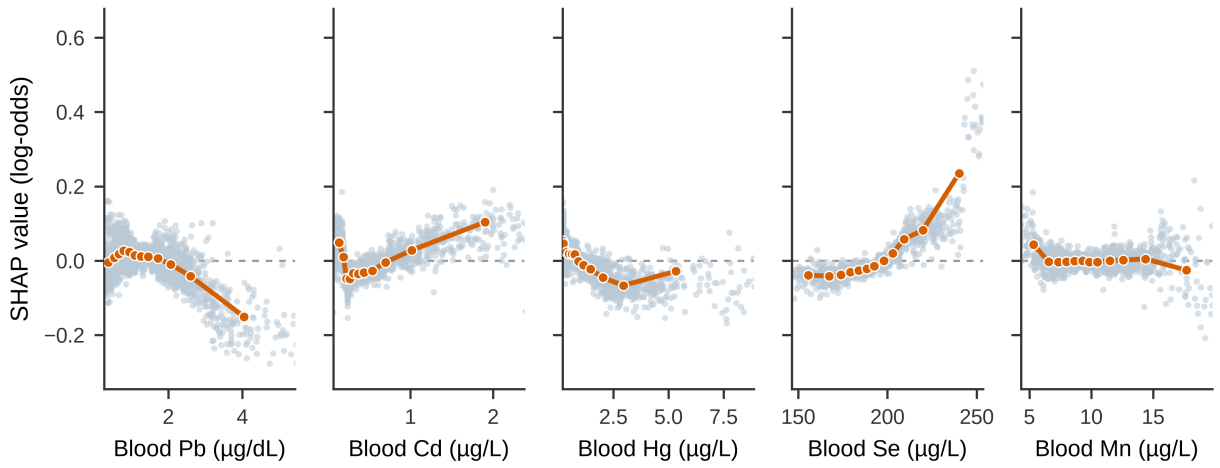


Figure 5. SHAP dependence for the five blood metals in the CatBoost model on the held-out test partition. Each grey point is one participant; the orange line joins mean SHAP values within duodeciles of the exposure distribution, and the dashed line at zero marks no contribution to the predicted log-odds. Axes are truncated at the 1st and 98th percentiles. Selenium and manganese are based on the 1,309 test participants from cycles in which those analytes were assayed. Abbreviations as in Figure 3.

5. Subgroup performance and sensitivity analyses

Discrimination was stable across sex and race/ethnicity but deteriorated markedly with age (Table S6, Figure S5). Against an overall test-set AUROC of 0.857 (0.840–0.872), the oldest tertile reached 0.741 (0.701–0.783), an interval lying entirely below that of the youngest tertile (0.861, 0.834–0.888). Within a single age tertile the range of age is compressed to about two decades, so the model's second most important predictor can no longer separate participants within that stratum, and a prevalence of 72.7% leaves few negatives to rank against. Calibration behaved differently: only the middle tertile departed detectably from ideal on both statistics (slope 1.19, 1.018–1.388; observed-to-expected 0.902, 0.845–0.964), with women showing a smaller over-prediction (0.943, 0.900–0.993); in the oldest tertile predicted risk was accurate in aggregate (1.011, 0.971–1.052). These are descriptive screens over 22 intervals without multiplicity adjustment. The model is therefore least useful for ranking precisely the group with the highest absolute burden, while still stating that group's average risk correctly.

Results of the four prespecified sensitivity analyses are given in Table S7. Adding physician-diagnosed diabetes raised cross-validated discrimination by 0.0099 to 0.0128 AUROC units, as expected for a variable partly downstream of the glucose component. Replacing SMOTENC with class weighting changed discrimination by at most 0.0010 units, confirming that the imbalance strategy is immaterial at this near-balanced prevalence. Substituting the measured-value-only outcome definition lowered discrimination by 0.0289 to 0.0338 units, the

largest effect of the four. The metal-panel analysis produced the most consequential result: restricting the panel to blood lead alone and discarding cadmium, mercury, selenium and manganese changed discrimination by between +0.0002 and +0.0011 units (marginally upward in all three models, not downward). Deltas this small sit within run-to-run variability of the cross-validation and carry no confidence intervals, so this is an absence of detectable contribution rather than a demonstration of exactly zero. The four additional metals add nothing measurable to discrimination: the ranking positions they occupy reflect within-model attribution, not predictive value that would be lost by removing them. No arm removed lead itself, so its own incremental contribution was not tested by this design.